

Assignment 11

$$\begin{aligned}u_F(w) &= u_G(w) = \sqrt{w} \\ \bar{u}_F &= 1 \\ \bar{u}_G &= 10 \\ V(w) &= -w\end{aligned}$$

$$\begin{aligned}P_F(Rain) &= \frac{2}{100} \\ P_G(Rain) &= \frac{20}{100}\end{aligned}$$

a)

$$\begin{aligned}\max_{w_R, w_{nR}} \quad & -w_R \left(\frac{20}{100} \right) - \frac{80}{100} \cdot w_{nR} \\ \text{s.t.} \quad & \\ \frac{1}{5} \cdot \sqrt{w_R} + \frac{4}{5} \cdot \sqrt{w_{nR}} & \geq 10 \quad (1) \\ \frac{1}{50} \cdot \sqrt{w_R} + \frac{49}{50} \cdot \sqrt{w_{nR}} & \leq 1 \quad (2)\end{aligned}$$

b)

we argue that both inequalities are binding

assume (1) is slack

$$\frac{1}{5} \cdot \sqrt{w_R} + \frac{4}{5} \cdot \sqrt{w_{nR}} > 10$$

decrease one or both wages by some $\epsilon, \delta \geq 0 : \epsilon + \delta > 0$ such that

$$\frac{1}{5} \cdot \sqrt{w_R - \delta} + \frac{4}{5} \cdot \sqrt{w_{nR} - \epsilon} > 10$$

And (2) of course still holds

This increases major's utility by $\frac{1}{5}\delta + \frac{4}{5}\epsilon > 0$

Meaning that this cannot be an optimal solution

assume

$$\begin{aligned}\frac{1}{50} \cdot \sqrt{w_R} + \frac{49}{50} \cdot \sqrt{w_{nR}} & < 1 \\ \frac{1}{5} \cdot \sqrt{w_R} + \frac{4}{5} \cdot \sqrt{w_{nR}} & = 10\end{aligned}$$

We do some math

$$\begin{aligned}\frac{1}{5} \cdot \sqrt{w_R} + \frac{4}{5} \cdot \sqrt{w_{nR}} & = 10 \Leftrightarrow \sqrt{w_R} = 50 - 4\sqrt{w_{nR}} \\ \frac{1}{50} \cdot \sqrt{w_R} + \frac{49}{50} \cdot \sqrt{w_{nR}} & = \frac{1}{50} \cdot (50 - 4\sqrt{w_{nR}}) + \frac{49}{50} \cdot \sqrt{w_{nR}} = 1 + \frac{45}{50} \cdot \sqrt{w_{nR}} < 1\end{aligned}$$

Which is not possible thus we must have

$$\begin{aligned}\frac{1}{50} \cdot \sqrt{w_R} + \frac{49}{50} \cdot \sqrt{w_{nR}} & = 1 \\ \Rightarrow w_{nR} & = 0\end{aligned}$$

giving us the contract

$$(w_R, w_{nR}) = (2500, 0)$$

c)

$$\max_{w_R, w_{nR}} -w_R \left(\frac{20}{100} \right) - \frac{80}{100} \cdot w_{nR}$$

s. t

$$\frac{1}{5} \cdot \sqrt{w_R} + \frac{4}{5} \cdot \sqrt{w_{nR}} \geq 10 \quad (1)$$

(1) must be binding otherwise decrease wage by $\epsilon > 0$

$$\frac{1}{5} \cdot \sqrt{w_R} + \frac{4}{5} \cdot \sqrt{w_{nR}} = 10 \Leftrightarrow \sqrt{w_R} = 50 - 4\sqrt{w_{nR}} \Leftrightarrow w_R = (50 - 4\sqrt{w_{nR}})^2$$

$$\max_{w_{nR}} -\frac{1}{5} (50 - 4\sqrt{w_{nR}})^2 - \frac{4}{5} \cdot w_{nR}$$

$$\frac{\partial V}{\partial w_{nR}} = -\frac{2}{5} (50 - 4\sqrt{w_{nR}}) \cdot -\frac{4}{2 \cdot \sqrt{w_{nR}}} - \frac{4}{5} = \frac{40}{\sqrt{w_{nR}}} - 4 = 0$$

$$\Leftrightarrow w_{nR} = 100$$

$$\Rightarrow w_R = 100$$

$$\frac{\partial^2 V}{\partial w_{nR}^2} = -\frac{20}{w_{nR}^{\frac{3}{2}}} < 0$$

Thus maximum

d) $w_R = w_{nR}$ is going to be optimal (first best)

$$\max_{w_R, w_{nR}} -w_R \left(\frac{20}{100} \right) - \frac{80}{100} \cdot w_{nR}$$

s. t

$$\frac{1}{5} \cdot \sqrt{w_R} + \frac{4}{5} \cdot \sqrt{w_{nR}} \geq \bar{u}_G \quad (1)$$

$$\frac{1}{50} \cdot \sqrt{w_R} + \frac{49}{50} \cdot \sqrt{w_{nR}} \leq \bar{u}_F \quad (2)$$

with

$$\bar{u}_G < \bar{u}_F$$

Show that the solution from the problem which is less constrained is still viable and thus optimal

$$w_R = w_{nR} = \bar{u}_G^2 \quad (\text{First Best solution})$$

$$\Rightarrow \frac{1}{50} \cdot \sqrt{w_R} + \frac{49}{50} \cdot \sqrt{w_{nR}} = \sqrt{w_R} = \bar{u}_G < \bar{u}_F$$

Thus this solution is optimal